

3.6 – 3.12

3.6 In each case, multiply out to obtain a sum of products: (Simplify where possible.)

a. $(W + X' + Z')(W' + Y')(W' + X + Z')(W + X')(W + Y + Z)$

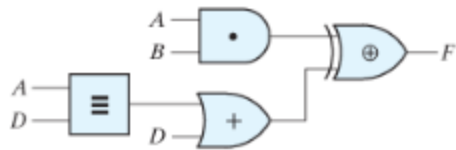
b. $(A + B + C + D)(A' + B' + C + D')(A' + C)(A + D)(B + C + D)$

3.7 Factor to obtain a product of sums. (Simplify where possible.)

a. $BCD + C'D' + B'C'D + CD$

b. $A'C'D + ABD' + A'CD + B'D$

3.8 Write an expression for F and simplify.



3.9 Is the following distributive law valid? $A \oplus BC = (A \oplus B)(A \oplus C)$ Prove your answer.

3.10 a. Reduce to a minimum sum of products (three terms):

$$(X + W)(Y \oplus Z) + XW'$$

b. Reduce to a minimum sum of products (four terms):

$$(A \oplus BC) + BD + ACD$$

c. Reduce to a minimum product of sums (three terms):

$$(A' + C' + D')(A' + B + C')(A + B + D)(A + C + D)$$

3.11 Simplify algebraically to a minimum sum of products (five terms):

$$(A + B' + C + E')(A + B' + D' + E)(B' + C' + D' + E')$$

3.12 Prove algebraically that the following equation is valid:

$$A'CD'E + A'B'D' + ABCE + ABD = A'B'D' + ABD + BCD'E$$